

Data Science 6

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Lecture 8

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Histograms, Summary Stats & more

0.1 Encoding

Definition 0.1. An **encoding** is a mapping from a variable to a visual graph/element.

Areas can be representative of the variable, not scaling but adding. Certain things are more effective for encoding by variable type. For categorical using **color hue, shape, and position(grouping)** allows us to avoid implying ordering as well.

0.2 Distributions

For numerical data, we use binning where $[a, b)$ unless it's the last bin. You can use **.bin** method to figure out what bins you could use. Histograms follow the area property which means

$$\text{area of bar} = \text{height of bar} \times \text{width of bin}$$

The area of the bars also must sum to 1 aka 100%.

Definition 0.2. Density: percent relative to amount of space

aka how crowded something is or. Look at like the normal distribution/bell curve.

Note. When using different bin sizes our y axis becomes density instead of count

Recall **.take** takes an array of row values and returns it and **np.arange** can give you list of numbers.

0.3 Percentiles

If we take a sorted version of the array. Reduce down percentile into $\frac{\text{rank}}{n}$ and choose the rank-nth element

0.4 Box Plots

Stats bruv all over.

0.4.1 Stats

Nah bro. Recall **range** and **IQR**. Also you can think the center of the data, and we have measures of center like mean, median. Variability is like how spread out the data is like **standard deviation**.

Note. Something cool about mean is that it's actually weighted by the number of times a value appears. So if each value appears twice in an array it's the same mean if they all appeared only once.

introduction to data analysis: Box Plot

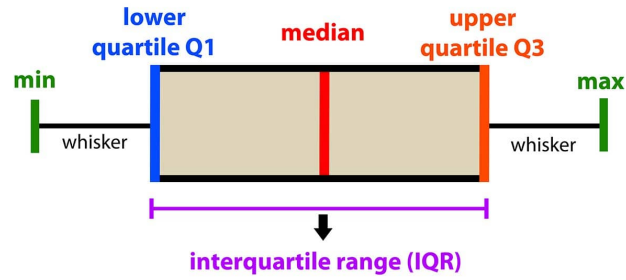


Figure 1: Box plot.

0.5 Grouping

The `.group` method returns the count that each row value appears in the given column. Can also pass in a function that is applied.

*Note. **Simpson's Paradox** in data reading essentially is where some trend or relationship disappears when we categorize data by certain groups. This is usually due to confounding variable.*